AI-Powered Portfolio Management System

Project Documentation

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1. Introduction

This project is my personal portfolio website. I made it with React and Firebase. It has three parts: the public website that visitors see, an admin panel where I manage all the content, and a chatbot called Santosh AI that answers questions about me.

I made it to show my skills, projects, experience and certificates to recruiters in one place. I also wanted to update the website easily, without changing the code every time I add a new project or certificate. That is why I built the admin panel.

I call it AI-Powered because the website has a chatbot that uses AI models to answer questions.

2. Objectives

- Show my skills, projects, experience, education and certificates in one place
- Update all the content from an admin panel without editing the code
- Let visitors send me a message through a safe contact form
- Let visitors ask questions about me to a chatbot
- Keep the admin panel safe from other people
- Make the website work well on phone, tablet and laptop screens

3. Project Overview

Part	Who uses it	What it does
Portfolio website	Visitors and recruiters	Shows my profile, skills, projects, experience, education and certificates. Has a contact form and a resume download button.
Admin panel	Only me	Used to add, edit and delete all the content, read messages and check security alerts.
Santosh AI chatbot	Visitors	Answers questions about my skills, projects, education, experience and certificates.

4. Technologies Used

Technology	Used for
React.js and Vite	Building the website and the admin panel
Tailwind CSS	Styling and responsive layout
Framer Motion	Small animations
React Icons	Icons on the website
Firebase Firestore	Database for all the portfolio content and messages
Firebase Authentication	Admin login

Technology	Used for
Cloudinary	Storing images, PDF files and PPT files
Vercel	Hosting the website and running the serverless functions
Firebase Admin (in serverless functions)	Safe server side actions like deleting messages
Resend	Sending email notifications
Cloudflare Turnstile	Stopping bots on the contact form
Gemini API	Main AI model for the chatbot
Groq, Mistral and OpenRouter	Backup AI models when Gemini does not work

5. How the System Works

1. All the portfolio content (hero text, about, skills, projects, certificates and more) is saved in Firebase Firestore.
2. The website reads this content from Firestore and shows it to the visitor. Because of this, a change in the admin panel shows on the website without any code change.
3. I log in to the admin panel and add or change the content. Files like images, PDF and PPT are uploaded to Cloudinary, and the file link is saved in Firestore.
4. The chatbot, the contact form and the admin security work through Vercel serverless functions. This keeps the secret keys outside the website code, so visitors cannot see them.

6. Portfolio Website

The website has these sections:

Section	What it shows
Home	My name, role, a short line about me, my Open to Work status and location, social links and the chatbot button.
About	A short introduction, role, location and small numbers like projects, certificates, CGPA and experience.
Skills	Skills in groups like Data Analytics and Web Development, with icons.
Projects	Project cards with search, a Featured / All filter and tech tags. The View Details popup shows the description, result, key highlights, tech stack and buttons for GitHub, Dashboard, View Deck and Download PPT.
Experience	My work experience.
Education	My education details.
Certificates	Certificate cards with search, a featured filter, type and skill tags, and a View Certificate link.
Contact	A contact form and my social links in the footer.

The numbers in the About section (projects and certificates) are counted automatically from the data in Firebase.

7. Admin Panel

The admin panel is private. Only I can log in. These are its modules:

Module	What it does
Dashboard	Shows the number of projects, certificates, experience and skills, with quick buttons to add or update content.
Hero	Update my role text, photo, Open to Work status and location. The photo can be cropped before upload.
About	Update the About text and details.
Skills	Add, edit and delete skills with a category.
Projects	Add, edit and delete projects. Each project can have a month and year, type, result, key highlights, tech tags, GitHub link, dashboard image, deck PDF and PPT file. A star button marks it as Featured.
Certificates	Add, edit and delete certificates with platform, type, skill tags and the certificate file. A star button marks it as Featured.
Experience and Education	Add, edit and delete these entries.
Resume	Upload, replace or delete my resume PDF.
Messages	Shows the messages from the contact form in Inbox and Spam. I can mark a message as Pending or Replied, mark it as unread, or delete it.
Security Alerts	Shows wrong login attempts and lets me block or unblock an IP address.
Notification bell	Shows new messages and new alerts in one place.

When I replace or delete a file, the old file is also removed from Cloudinary, so no unused files stay there.

8. Santosh AI Chatbot

Santosh AI is a small chatbot on the website. It answers questions about me, such as my skills, projects, education, experience, certificates, contact details and resume. This is how it answers a question:

1. The visitor types a question.
2. If the question is about a project, skill, certificate, education, experience, contact or resume, the answer is made straight from my data in Firebase. No AI model is used here, so it is fast and it does not use the daily limit.
3. For other questions about me, the chatbot sends the question to an AI model. It tries Gemini first. If Gemini does not work, it tries Groq, then Mistral, then OpenRouter.

4 If the question is not about me, the chatbot says politely that it can only answer questions about me.

5 If a visitor has already asked 15 AI questions in one day, the chatbot shows a friendly message and suggests using the contact form.

Popular answers are saved for some time so the replies come faster.

9. Contact Form

Feature	Details
Fields	Name, email, subject, website link (optional), message (up to 1000 characters) and up to 3 attachments (JPG, PNG, WebP or PDF, 5 MB each).
Bot protection	Cloudflare Turnstile is used to stop bots.
Spam folder	Messages that look like spam are saved in a separate Spam folder in the admin panel. They are not deleted.
Duplicate messages	The same message from the same email within 24 hours is blocked.
Daily limit	There is a daily limit of messages for one email.
Junk messages	Very short or junk messages like abcd are rejected.
Messages for visitors	The visitor sees a clear message for success and for each type of error.
Email notification	I get an email when a real message comes.

10. Security

Since the admin panel can change my whole website, I added some safety steps:

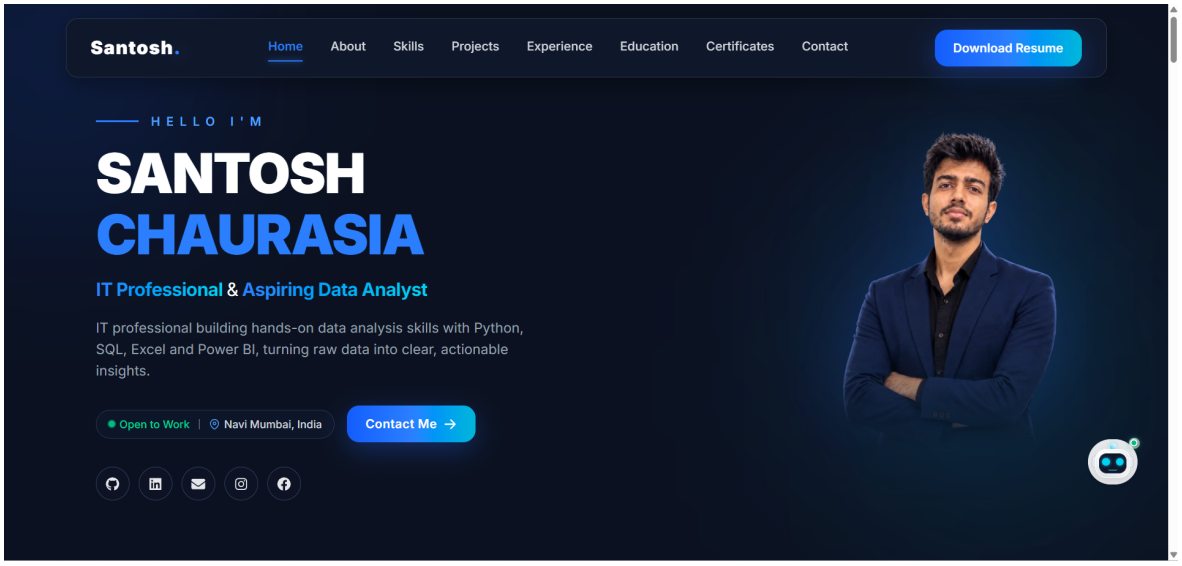
- Admin login needs a username or email with a password, and then a code from an authenticator app
- After some wrong login attempts, the IP address is blocked for some time and I get an alert
- The Security Alerts page shows these attempts, and I can block or unblock an IP address myself
- Firestore rules allow only the admin account to change the data
- Secret keys are kept on the server side and not in the website code
- The contact form uses Turnstile, spam checks and message limits

11. Responsive Design

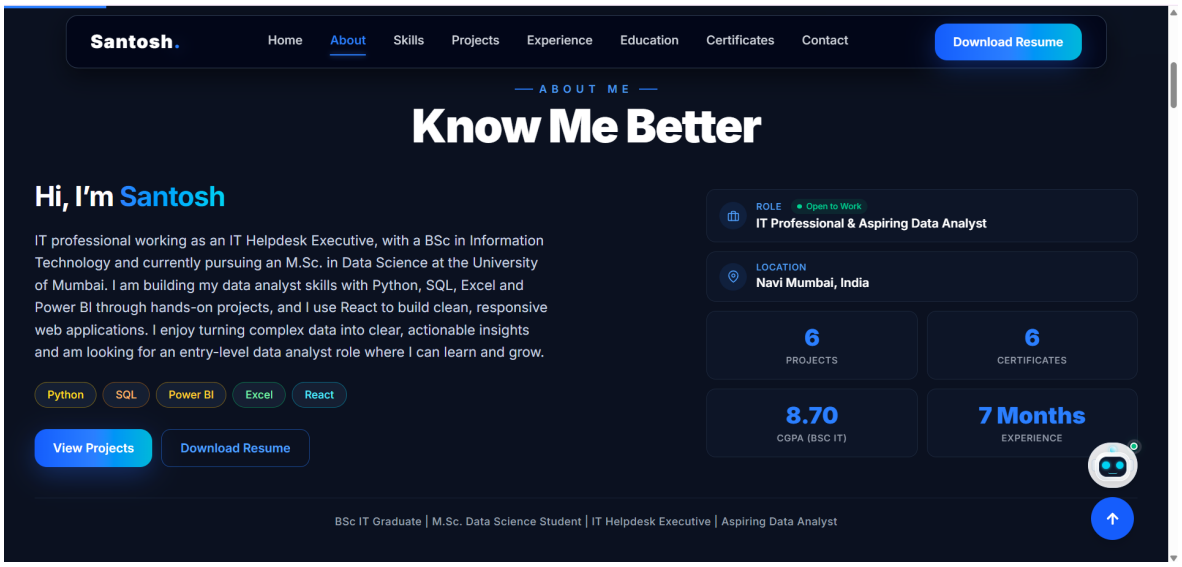
I made the website responsive so it looks good on different devices. I checked every page on these 7 screen sizes:

No.	Size	Device type
1	375 x 667	Small phone
2	640 x 900	Large phone
3	768 x 1024	Tablet
4	1024 x 768	Small laptop
5	1280 x 800	Laptop
6	1536 x 900	Large laptop
7	1920 x 1080	Full HD screen

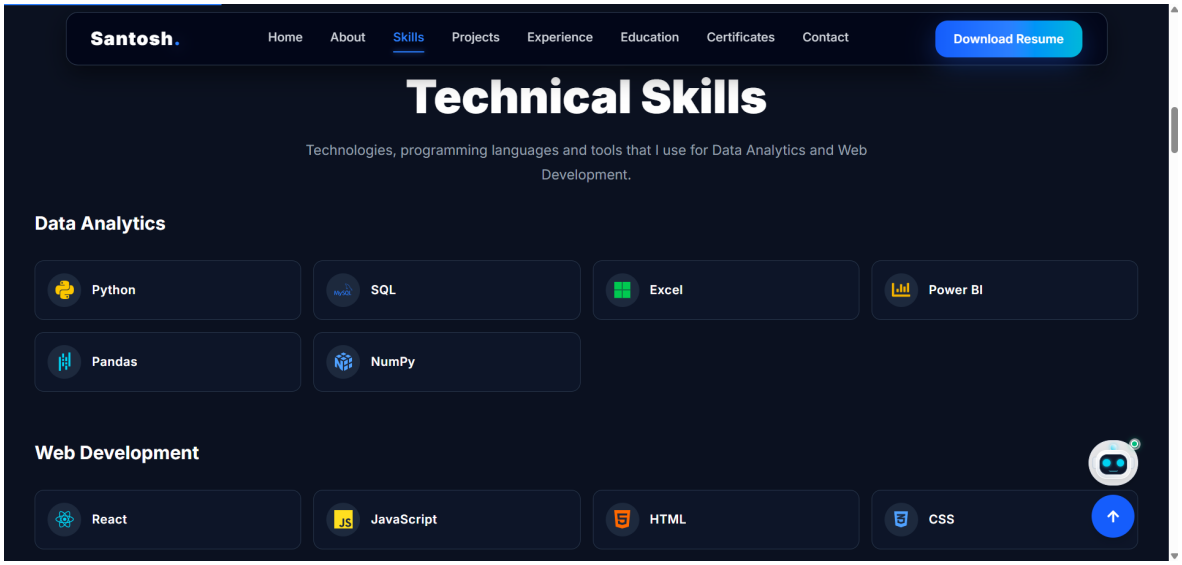
12. Screenshots



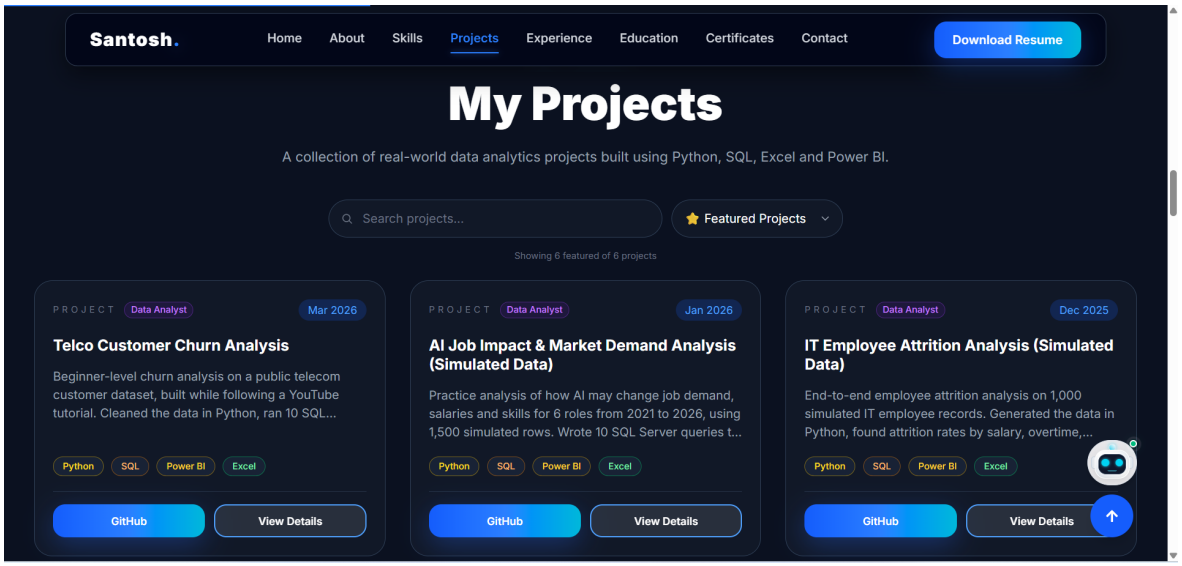
Home page



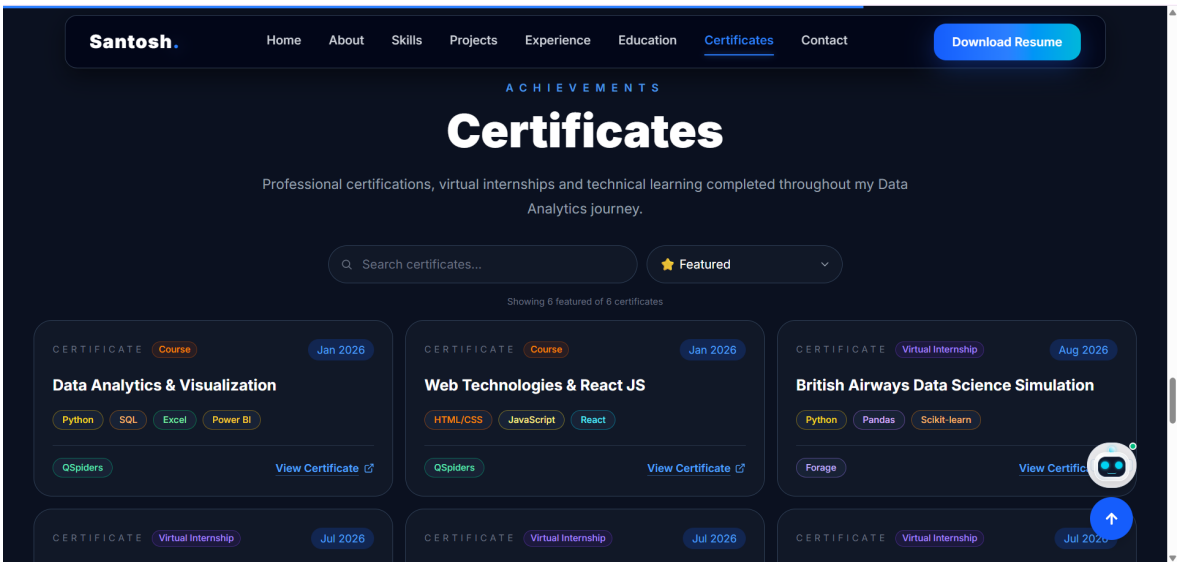
About section



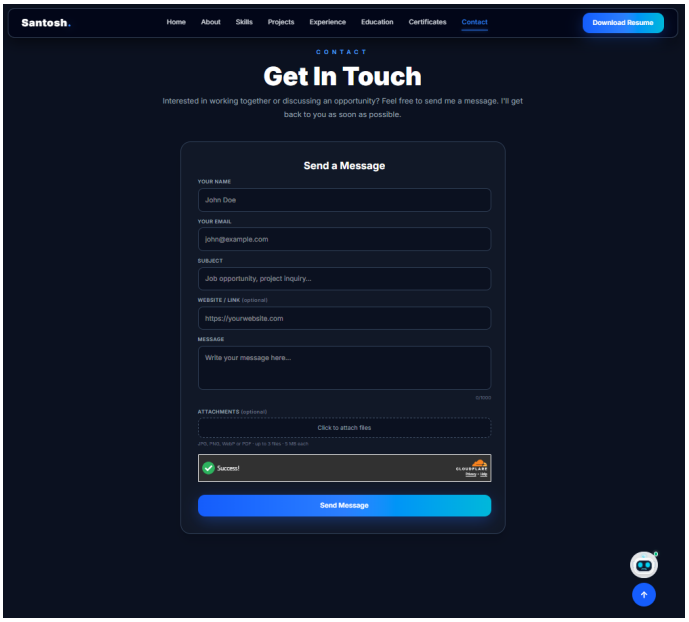
Skills section



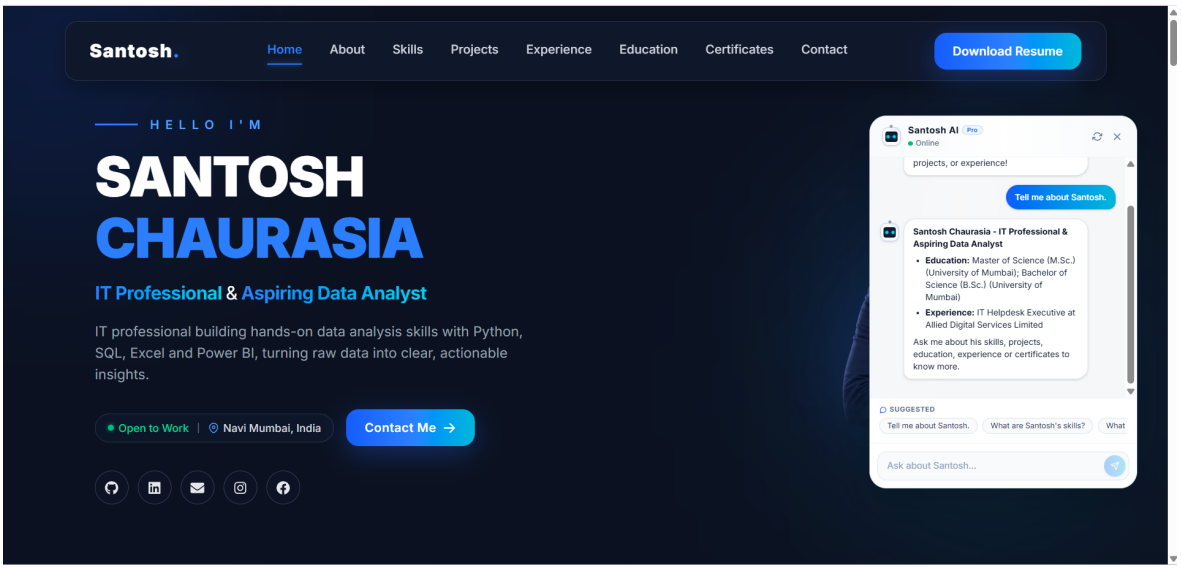
Projects section



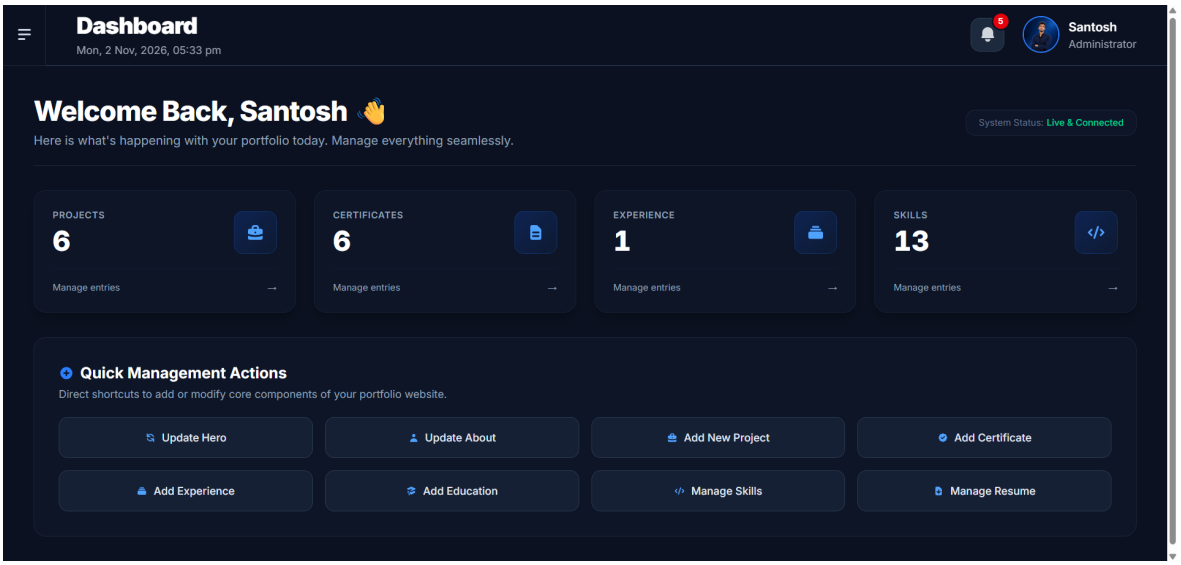
Certificates section



Contact form



Santosh AI chatbot



Admin dashboard

13. Problems I Faced and How I Solved Them

Problem	What I did
The project popup was cut on small phones.	I changed the layout. The top part and the buttons stay fixed, and only the middle part scrolls.
Spam messages in the contact form.	I added Turnstile, a Spam folder, a duplicate check and a daily limit.
The free Gemini quota can finish.	I added backup models (Groq, Mistral, OpenRouter) and a daily limit of 15 AI questions per visitor.
Old files stayed in Cloudinary after I replaced them.	I added automatic deleting of the old file when a file is replaced or removed.
A server function crashed after I deployed it.	I found that the problem was in the module setup. I fixed it by checking the login token in a simpler way.

Problem	What I did
The Hero photo moved away from the text on very wide screens.	I put the photo and the text inside one container with a maximum width.
The dashboard image in the project popup looked blurry.	I used a higher quality setting for the image and I use bigger screenshots now.

14. Testing

- Checked all the pages on the 7 screen sizes listed above
- Tested the contact form on the live website with different cases, like a normal message, a duplicate and a junk message
- Tested deleting messages and replacing or deleting files on the live website
- Tested the admin login steps and the Security Alerts page on the live website
- Tested the project popup with a dashboard image, a deck PDF and a PPT file

15. Limitations

- The project runs on free plans, so limits can come, like the AI quota and the daily message limit
- The chatbot knows only the data of my portfolio, and an AI reply may not always be perfect
- The admin panel is made for one user only, which is me
- The source code is private, so this repository has only the details and screenshots

16. Future Scope

- Add light animations on more sections
- Improve SEO so the website shows up in Google search

17. Conclusion

This project helped me learn how a complete web application is built, from the database and the login to file uploads, serverless functions and AI models. It also gave me a real website that I use to show my work to recruiters. I will keep improving it with new features.

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